ANGULAR ANOMALIES IN PARAMETRIC ELLIPSES

Annotation. The paper presents the results of a computational experiment studying the dependence of angles θ_1 and θ_2 on the parameters of length (L) and height (h). The experiment was performed for values of $\lambda = L/h$ in the range 6 – 21.64. The data were obtained using the ATAN2 functions in Excel, taking into account trigonometric transformations. The results demonstrate the nonlinear dynamics of the angles with a change in λ, which indicates the critical influence of the ratio of geometric parameters.

Keywords: angular anomalies, parametric ellipses, nonlinear dynamics.

Introduction.

A random experiment

It all started with a routine job in Microsoft Excel. I built ellipses using classical formulas, checking geometric models.

But when I entered the data for the parameter L (half-axis length) and angle θ, the calculation column returned something inexplicable:

At L=3.5, the angle was stable at 340.5°.

At L=4.14, it was abruptly divided into 149° and 211°.

And at L = 10.82, it dropped to 6° altogether.

"Excel bugs?" I thought. But the anomalies were real.

A pattern that cannot be ignored.

Despite the lack of mathematical justification, I have recorded clear patterns (Table 1, graph 1, graph 2):

Stability at low L:

At L <3.5, the angle is always 340.5°.

It's as if the system is "stuck" in its initial state.

Critical point L = 4.14:

The values branched by 149° and 211° without intermediate options.

"It looks like a phase transition in physics - but in spreadsheets?!"

The stabilization zone:

At L> 4.14, the angles were grouped around 180° (170°, 172°, 188°).

Decay anomaly:

At L =10.82, the angle of 6° violated all expectations.

"It resembled quantum laws, but it arose in a simple geometric model. I called them Ovchinnikov's Laws.

Hypothesis 1. An Excel bug?

I have tested all possible error causes.:

Formulas: =ATAN2(...)*180/PI () is correct.

Rounding: Increasing the precision to 15 digits did not help.

Hardware failures: Recalculation on another PC gave identical results.

Conclusion. This is not a technical error. The data is reproducible with an accuracy of 0.1°.

So it's either an Excel error, or:

Hypothesis 2:

New physical laws - Ovchinnikov's Laws

1. At $\lambda = L/0.5$:

At high L/h, the spiral tends to a straight line ($\theta = 180°$).

At $\lambda \leq 5.0 \rightarrow \theta \equiv 340.5°$, there is a singularity.

Angle: $\theta \equiv 340.5°$.

The system ignores changes in the L-energy barrier???

The system is "frozen" in its initial state. There is not enough energy to overcome the potential barrier.

$5.0 \leq \lambda \leq 7.0$ – linear decline

$\theta = 340.5 - 70.5 \times (7-\lambda)/2$ (1)

2. Bifurcation ($\lambda = 8.28$).

At $7.0 \leq \lambda \leq 8.28$– the transition to bifurcation.

Critical value: $\lambda = 8.28$.

Angles: Two stable states - 149° (lower) and 211° (upper).

Bifurcation of the angle - quantum tunneling???

Assembly disaster. The slightest disturbance picks a branch.

The bifurcation point ($\lambda = 8.28$, $\theta \in (149°, 211°)$).

3. Stabilization ($\lambda > 8.28$), at $\lambda > 8.28 \rightarrow \theta \rightarrow 180° \pm \delta(\lambda)$.

At $\lambda \in [8,0, 20]$

Angles: 170° - 188°

$\theta = 180 \pm 31 \times e^{(-0.15(\lambda - 8.28))}$ (2)

Exponential tendency to 180° with fluctuations.

Angles in physical systems are quantized by discrete values (340.5°, 149°, 211°, etc.).

4. Decay anomaly ($\lambda = 21.64$; $\theta = 6°$)

At $\lambda > 20$.

$\theta = 6 \pm 174 \times e^{(-0.25(\lambda - 20))}$ (3)

"Value 6°" new state of matter???

During repeated calculations with a step of $\Delta L = 0.01$, the patterns were confirmed.

"The system behaved like it was alive: it resisted changes, then abruptly "switched", and finally "died" at a critical stretch. But why? Mathematics was silent"

Discussion

The dilemma, where to look for the truth?

I have formulated questions for the scientific community.

Theoretical physics:

Can angular anomalies be a manifestation of nonlinear dynamics?

Are the jumps related to Volume disasters (folding, assembling)?

Materials Science:

Analogies to alloy deformation:

Ductility (stable angle) → Crack (bifurcation) → Fracture (disintegration).

Cosmology:

Parallels with the evolution of the Universe:

Singularity (340.5°) → Inflation (bifurcation) → Dark energy (6°).

"I am not a physicist or a mathematician. I just saw a pattern, and I'm asking you to help me decipher it."

Conclusion

Invitation to a dialogue

Fact: Anomalies exist and are reproducible.

Mystery: Their nature is unknown.

Perhaps it is:

1. An artifact of the model (but why is it correct for 90% of the points?),

2. The manifestation of quantum effects in the macrocosm, or... a new physics section, born in Excel.

"If you have found out these patterns, please contact me. If not, repeat the experiment:

Set h = 0.5, L = [3.0, 10.82].

Find your 340.5°, 149° and 6°.

Mathematics is the language of the universe. Perhaps we have heard her new word"

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